

1. If a relation on $A = \{a, b, c\}$ is $R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$, then R is:

- **Reflexive:** Because $(a, a), (b, b), (c, c) \in R$ ✓
 - **Symmetric:** Because $(a, b) \in R \Rightarrow (b, a) \in R$ ✓
 - **Transitive?** Check $(a, b), (b, a) \in R \Rightarrow (a, a) \in R$ ✓, also all other pairs satisfy.
Answer: Reflexive and Symmetric only (c).
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2. Domain of $f(x) = 4 - x^2$

- This is a polynomial, and **polynomials are defined for all real numbers.**
Answer: R (c).
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3. If $f(x) = \frac{1}{x}$ and $g(x) = x^2$, domain of $f \circ g$

- $f \circ g(x) = f(g(x)) = \frac{1}{x^2}$
- $x^2 \neq 0 \Rightarrow x \neq 0$

Answer: All real except 0 (c). But your key says a, which seems incorrect; double-checking: domain is $x \in \mathbb{R} \setminus \{0\}$.

4. Which function is bijective over R?

- **Bijective:** both one-one (injective) and onto (surjective)
 - $f(x) = x^2 \rightarrow$ Not injective (fails on negative x)
 - $f(x) = |x| \rightarrow$ Not injective
 - $f(x) = x^3 \rightarrow$ Injective and surjective ✓
 - $f(x) = e^x \rightarrow$ Injective but not surjective onto all real numbers
Answer: $f(x) = x^3$ (c).
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5. If $f(x) = |x - 2|$, range:

- Absolute value is always ≥ 0 , so **range** = $[0, \infty)$
Answer: a
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6. Number of equivalence relations on a set with 3 elements:

- Number of **partitions of a set** = 5
Answer: a
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7. If f is injective:

- One-one function 

Answer: a

8. Number of invertible functions from A to B if $|A| = 4$

- Invertible functions exist only if $|A| = |B|$, so 4! permutations

Answer: 24 (c)

9. Range of $f(x) = x^2, x \geq 0$:

- $x^2 \geq 0 \rightarrow \text{range} = [0, \infty)$

Answer: a

10. For what values of a is $f(x) = ax + 2$ invertible?

- Linear function is invertible if $a \neq 0$

Answer: c

11. Domain of $f(x) = \sqrt{x}$

- $x \geq 0$

Answer: c

12. Function $f(x) = x^3$ is:

- Injective and surjective $\rightarrow$ **bijective**

Answer: c

13. Range of $f(x) = x^2 - 4$

- $x^2 \geq 0 \Rightarrow x^2 - 4 \geq -4 \rightarrow \text{range} = [-4, \infty)$

Answer: a

14. If f is even:

- $f(-x) = f(x)$ 
- Symmetric about **y-axis**, not x-axis

Answer: a

15. $f(x) = |\sin x|$

- Repeats every $\pi \rightarrow$ **periodic**

Answer: b

16. Number of functions from $A = \{1, 2, 3\}$ to $B = \{1, 2\}$

- Each element in A has 2 choices $\rightarrow 2^3 = 8$

Answer: b

17. Domain of $f(x) = \ln(\ln x)$

- $\ln x > 0 \rightarrow x > 1$

Answer: a

18. Range of $f(x) = \ln(\ln x)$

- $\ln(\ln x) \rightarrow$ All real numbers possible as $\ln y \in \mathbb{R}$

Answer: b ($\mathbb{R}$)

19. If f and g are bijective, then $f \circ g$ is:

- Composition of bijections $\rightarrow$ **bijective**

Answer: c

20. Domain where $f(x) = x^2$ is invertible

- $f(x) = x^2$ is invertible only on $x \geq 0$ or $x \leq 0$
 - Answer: $x \geq 0$ (b)
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21. Invertible functions are always:

- Must be bijective 

Answer: c

22. Range of $f(x) = \tan^{-1} x$

- $\tan^{-1} x \in (-\pi/2, \pi/2)$

Answer: b

23. Domain of $f(x) = 1/(x - 1)$

- Denominator $\neq 0 \rightarrow x \neq 1$

Answer: a

24. If R is symmetric and transitive but not reflexive, add:

- To make equivalence, add all (a,a)

Answer: a

25. Domain of $f(x) = \ln(x + 2)$

- $x + 2 > 0 \rightarrow x > -2$

Answer: a

26. Range of $f(x) = x^2 - 4$

- Minimum value -4 $\rightarrow$ range = $[-4, \infty)$

Answer: b

27. Number of reflexive relations on set with 3 elements:

- Reflexive relations: $2^{(n^2-n)} = 2^{(3^2 - 3)} = 2^6 = 64$

Answer: b

28. "Divides" relation on positive integers:

- Reflexive ☒, Transitive ☒, Antisymmetric ☒

Answer: d

29. Domain of $f(x) = \sqrt{x - 1}$

- $x - 1 \geq 0 \rightarrow x \geq 1$

Answer: a

30. Range of $f(x) = x^2 - 4$ (repeated)

- $[-4, \infty)$

Answer: c

31. If f is odd:

- $f(-x) = -f(x)$ ✓

Answer: b

32. Range of $f(x) = 2x/\sqrt{1+x^2}$

- Max/min: $[-1, 1]$ → cannot be ± 1 exactly? Check → range all except ± 1
 - Answer: all except 1 → b
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33. Domain of $f(x) = \sin^{-1}(3x)$

- $|3x| \leq 1 \rightarrow |x| \leq 1/3$

Answer: b

34. Range of $f(x) = \sin^{-1}(3x)$

- $\sin^{-1} \in [-\pi/2, \pi/2] \rightarrow$ for $x \in [-1/3, 1/3]$, range = $[-\pi/2, \pi/2]$ scaled?
 - Answer: b
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35. Domain of $f(x) = \log(x+2)$

- $x+2 > 0 \rightarrow x > -2$

Answer: a

36. Number of anti-symmetric relations on 2-element set:

- Formula: $3^{(n(n-1)/2)} * 2^n$
- For 2 elements: $3^1 * 2^2 = 3*4=12$? Actually, known result = 7

Answer: c

37. $f(x) = x^3+1$, inverse exists because f is:

- Strictly increasing → one-one

Answer: a

38. Inverse of matrix exists when:

- $ad - bc \neq 0$

Answer: b

39. If f is bijection, cardinality of f^{-1} possible is:

- Exactly 1 inverse function 

Answer: a

40. Range of $f(x) = x + x^1, x > 0$

- Simplifies $\rightarrow$ linear increasing $\rightarrow$ range = $(0, \infty)$

Answer: b